Final Exam

● Graded

Student

Matthew Schmidt

Total Points

41.5 / 45 pts

Question 1

Overall Grades **41.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 3.5 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Matthew Schmidt	mws57	Matthew Schmidt

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Matthew Schmidt

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	4.5	5	5	5	5	5	0[2]	41.5

Problem 1. [5] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

Gen = particular + Homogenous

since $n = 3$ & $m = 3$, 1 solution

$$A = \left[\begin{array}{ccc|c} 2 & -3 & 0 & 0 \\ 0 & -2 & -3 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right] \quad R_1 \leftarrow R_1 - 2R_3, \quad R_2 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & \\ 0 & -2 & -3 & \\ 0 & -3 & -4 & \end{array}\right] \quad R_2 \leftarrow R_2 - R_3$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & \\ 0 & -2 & -3 & \\ 1 & 0 & 2 & \end{array}\right] \quad R_1 \leftarrow R_1 - 2R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & \\ 0 & 1 & 1 & \\ 0 & -3 & -4 & \end{array}\right] \quad R_3 \leftarrow R_3 + 3R_2$$

$$\left[\begin{array}{ccc|c} 0 & -3 & -4 & -1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \quad R_3 \leftrightarrow R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & \\ 0 & 1 & 1 & \\ 0 & 0 & -1 & \end{array}\right] \quad R_3 \leftarrow -R_3$$

R

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right] \quad R_2 \leftarrow R_2 - R_3$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & -3 & -4 & -1 \end{array}\right] \quad R_3 \leftarrow R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 5 \end{array}\right] \quad R_3 \leftarrow -R_3$$

$22 - 21 = 1$

$-14 + 15 = 1$

$11 - 10 = 1$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right] \quad R_2 \leftarrow R_2 - R_3, \quad R_1 \leftarrow R_1 - 2R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$$

5/5

mxn = 4x3

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

1. Casting out Algorithm

$R_1 \leftarrow R_1 - 2R_4$

$R_2 \leftarrow R_2 + 5R_4$

$\begin{bmatrix} 0 & -2 & -2 \\ 0 & 15 & 15 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_4 \leftrightarrow R_1$

1. Basis for col(A)

$\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 15 & 15 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix}$ $R_2 \leftarrow R_2 - 15R_3$

$R_2 \leftrightarrow R_4$

2. Basis for row space row(A)

$\{[1\ 3\ 2], [0\ 1\ 1]\}$ DIC

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & -2 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $R_2 \leftarrow R_2 + 2R_3$

$R_2 \leftrightarrow R_3$

3. nul(A)

$\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

$\begin{bmatrix} \textcircled{1} & 3 & 2 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 2 pivots

$R_1 \leftarrow R_1 - 3R_2$

$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix}$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

4.5/5

$A^{*}_{2\times 3}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$A = \left[T(e_1), T(e_2), T(e_3)\right]$

$$A = \begin{bmatrix}-2 & -1 & -3\\ 0 & -1 & -5\end{bmatrix}$$

$a_1v_1 + a_2v_2 + a_3v_3 = e_1, e_2, e_3$

$$\left[\begin{array}{ccc|ccc} a_1 & a_2 & a_3 & e_1 & e_2 & e_3\\ 0 & 1 & 1 & 1 & 0 & 0\\ 1 & 0 & 0 & 0 & 1 & 0\\ -1 & -1 & -1 & 0 & 0 & 1\end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0\\ 1 & 0 & 0 & 0 & 1 & 0\\ 0 & -1 & -1 & 0 & 1 & 1\end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0\\ 0 & 1 & 1 & 1 & 0 & 0\\ 0 & 0 & 0 & -1 & -1 & 1\end{array}\right]$$

for $T(e_3)$: $a_1 = 0$ $a_2 = 1$ $a_3 = -1$

$$T(e_1) = T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix} - \begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = T\begin{pmatrix}1\\0\\-1\end{pmatrix} - T\begin{pmatrix}0\\0\\-1\end{pmatrix} = \begin{bmatrix}0\\2\end{bmatrix} - \begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}-2\\0\end{bmatrix}$$

$v_2 \quad v_3$

$T(e_2) = a_1 = 1 \quad a_2 = 0 \quad a_3 = -1$

$$= T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix} - \begin{bmatrix}0\\0\\-1\end{bmatrix}\right)$$

$$\begin{bmatrix}1\\1\end{bmatrix} - \begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}-1\\-1\end{bmatrix}$$

or

$T(e_3) = a_1 = -1 \quad a_2 = -1 \quad a_3 = -1$

$$T\left(-\begin{bmatrix}0\\1\\-1\end{bmatrix} - \begin{bmatrix}0\\1\\-1\end{bmatrix} - \begin{bmatrix}0\\0\\-1\end{bmatrix}\right)$$

$$-\begin{bmatrix}1\\1\end{bmatrix} - \begin{bmatrix}0\\2\end{bmatrix} - \begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}-3\\-5\end{bmatrix}$$

Problem 4. [5'] Find the Jordan canonical form of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

1. find λ
2. find geometric mults...
3. find P

$$J = \left[\begin{array}{cc:c} 1 & 1 & 0 \\ 0 & 1 & 0 \\ \hdashline 0 & 0 & 2 \end{array}\right]$$

$$\det \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{vmatrix}$$ expand on 3rd column

$$P = \begin{bmatrix} 0 & -1/2 & -2 \\ 0 & 0 & -1 \\ 1 & 0 & 6 \end{bmatrix}$$

$$(1-\lambda)\begin{vmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix}$$

$$(1-\lambda)(1-\lambda)(2-\lambda) = 0$$

where $A = P^{-1}JP$ or $PA = JP$

$\lambda = 1, 1, 2$ find $\gamma(\lambda_i)$

chain for $\lambda = 1$

$$\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{array}\right] \vec{x} = 0$$

$$\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 1 \end{array}\right] \quad R_3 \leftarrow R_3 + R_1$$

$$R_3 \leftarrow -\tfrac{1}{2} R_3$$

$$\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 0 & 1 \end{array}\right] \quad R_2 \leftarrow R_1 - 2R_2$$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \quad R_2 \leftrightarrow R_1$$

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 0 & 1 \end{array}\right] \quad R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \quad R_3 \leftarrow R_3 - 2R_2$$

$$R_1 \leftarrow R_1 \cdot -\tfrac{1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1/2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \leftarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} t$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1/2 \\ 0 \\ t \end{pmatrix} \text{ let } t = 0$$

solutions $\lambda = 2$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} -1/3 \\ -1/6 \\ 1 \end{pmatrix} \text{ let } t = 6 = \begin{pmatrix} -2 \\ -1 \\ 6 \end{pmatrix}$$

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

where $\vec{x} = P\vec{y}$ and $D = P^TAP$

$$A = \begin{matrix} & x & y & z \\ x & 3 & 0 & 1 \\ y & 0 & 3 & -1 \\ z & 1 & -1 & 4 \end{matrix}$$

$$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{22} & 1/\sqrt{11.5} \\ 1/\sqrt{2} & 1/\sqrt{22} & -3/\sqrt{11.5} \\ 1/\sqrt{2} & 0 & 3/\sqrt{11.5} \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$$\begin{vmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{vmatrix}$$

exp along 3rd column

$(3-\lambda)^3 -$

$A_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

or $\begin{vmatrix} 0 & 3-\lambda \\ 1 & -1 \end{vmatrix} + \begin{vmatrix} 3-\lambda & 0 \\ 1 & -1 \end{vmatrix} + (4-\lambda)\begin{vmatrix} 3-\lambda & 0 \\ 0 & 3-\lambda \end{vmatrix}$

$-(3-\lambda) + -(3-\lambda) + (4-\lambda)(3-\lambda)^2$

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

or $(3-\lambda)\begin{vmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 3-\lambda & -1 \end{vmatrix}$

$(3-\lambda)\left[(3-\lambda)(4-\lambda)-1\right] + -(3-\lambda)$

$\lambda^2 - 7\lambda + 12 - 1$

$(3-\lambda)\left((\lambda^2-7\lambda+11)\right)$

$-\lambda^3 + 7\lambda^2 - 11\lambda$

$+3\lambda^2 - 21\lambda + 33$

$-\lambda^3 + 10\lambda^2 - 32\lambda + 33 + \lambda - 3$

$-\lambda^3 + 10\lambda^2 - 31\lambda + 30 = 0$

$(\lambda - 2)$ is factor

$(\lambda-2)(-\lambda^2+8\lambda-15) = 0$

$(-\lambda+5)(\lambda-3)$

$\lambda = 2, 3, 5$

$A_5 = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix}$

$$\begin{bmatrix} 0 & -2 & -1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t\begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}$$

$\|\vec{v}\| = \sqrt{1 + \frac{1}{4} + 1} = \frac{3}{2}$

$\Rightarrow \vec{v}$ of distinct λ are all orthogonal (only need to normalize)

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \text{ and } \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

v_1 v_2 v_3 w

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \quad R_2 \leftarrow R_2 + R_2$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right] \quad R_2 \leftrightarrow R_1$$

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right] \quad \begin{array}{l} R_1 \leftarrow R_1 + R_3 \\ R_2 \leftarrow R_2 - 2R_3 \end{array}$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 2 & 8 \\ 0 & 0 & -3 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right] \quad R_2 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 2 & 8 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right] \quad R_3 \leftarrow R_3/-3$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 2 & 8 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right] \quad \begin{array}{l} R_2 \leftarrow R_2 - R_3 \\ R_1 \leftarrow R_1 - 2R_3 \end{array}$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right] \quad R_1 \leftarrow \frac{R_1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$a = 3$

$b = -1$

$c = 1$

$a =$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$v_2 \cdot v_1 = 2$

$v_1 \cdot v_1 = 2$

$v_1 \quad v_2$

1. Gram-Schmidt for orthogonal

$$\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}\right\}$$

$v_1 \quad u_2$

$u_2 = v_2 - proj_{v_1}(v_2)$

$$= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$- \frac{2}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$$

$$u_2 = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$$

2. Normalize

$\|u_1\| = \sqrt{2}$ $\|u_2\| = \sqrt{12}$

$$\left\{\begin{bmatrix}0\\1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix}, \begin{bmatrix}3/\sqrt{12}\\-1/\sqrt{12}\\1/\sqrt{12}\\-1/\sqrt{12}\end{bmatrix}\right\}$$

4/4

non invertible

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

skew symmetric

→ since $A^T = -A$, its skew symmetric

→ All entries on diagonal are 0

→ ~~possibly be reduced to RREF~~, since diagonals

At least for 3x3:

$$A = \begin{bmatrix} 0 & -i & -j \\ i & 0 & -k \\ j & k & 0 \end{bmatrix}$$

$$\det(A) = -i\begin{vmatrix} -i & -j \\ k & 0 \end{vmatrix} + j\begin{vmatrix} -i & -j \\ 0 & -k \end{vmatrix}$$

$$-ijk + jik = 0$$

so $\det(A^T) = \det(A)$

$\Downarrow$

$\det(-A) = \det A$

$\Downarrow$

$\det(-1A)\ \det(A)$

$\Downarrow$

$\det(kA) = k^n \det(A) \Rightarrow (-1)^n \det A = \det(A)$

only true for odd values of n is the det(A) factor is = 0

2/5

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

suppose $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \end{bmatrix}$ $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}$

dot product = 0 ✓

$\|\vec{u}+\vec{v}\| = \sqrt{(u_1+v_1)^2+(u_2+v_2)^2}$

$\|\vec{u}-\vec{v}\| = \sqrt{(u_1-v_1)^2+(u_2-v_2)^2}$

only works for $n=2$

if $\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$ then

$\sqrt{(u_1+v_1)^2+(u_2+v_2)^2} = \sqrt{(u_1-v_1)^2+(u_2-v_2)^2}$

- square both sides & multiply out

$u_1^2 + 2u_1v_1 + v_1^2 + u_2^2 + 2u_2v_2 + v_2^2 = u_1^2 - 2u_1v_1 + v_1^2 + u_2^2 - 2u_2v_2 + v_2^2$

cancel terms

$2u_1v_1 + 2u_2v_2 = -2u_1v_1 - 2u_2v_2$

$\Downarrow$

$4u_1v_1 + 4u_2v_2 = 0$

$\Downarrow$

$u_1v_1 + u_2v_2 = 0$

$\Downarrow$

$\vec{u}\cdot\vec{v} = 0$

$\Downarrow$

$\|\vec{u}\|\|\vec{v}\|\cos(\theta) = 0$

- $\cos = 0$ at $90°$ & $270°$ ($-90°$), so assuming $\|\vec{u}\|, \|\vec{v}\| \neq 0$,

they must be perpendicular.

(see back for attempt with size n vector)